Time Series Analysis: 780 (Tampere University)

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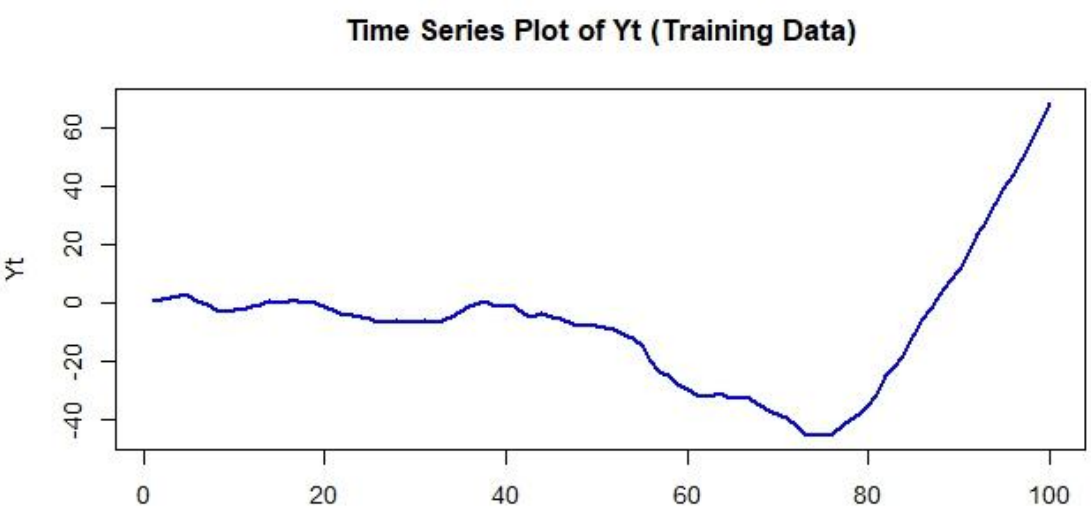
Column Assigned: V7

Assignment 1

Step 1: Preliminary analysis of Orders

1. Plot the times series (Yt)

```
ts_train <- ts(train)
plot(ts_train,  main="Time Series Plot of Yt (Training Data)", ylab="Yt", xlab="Time", col="blue",lwd=2)
```

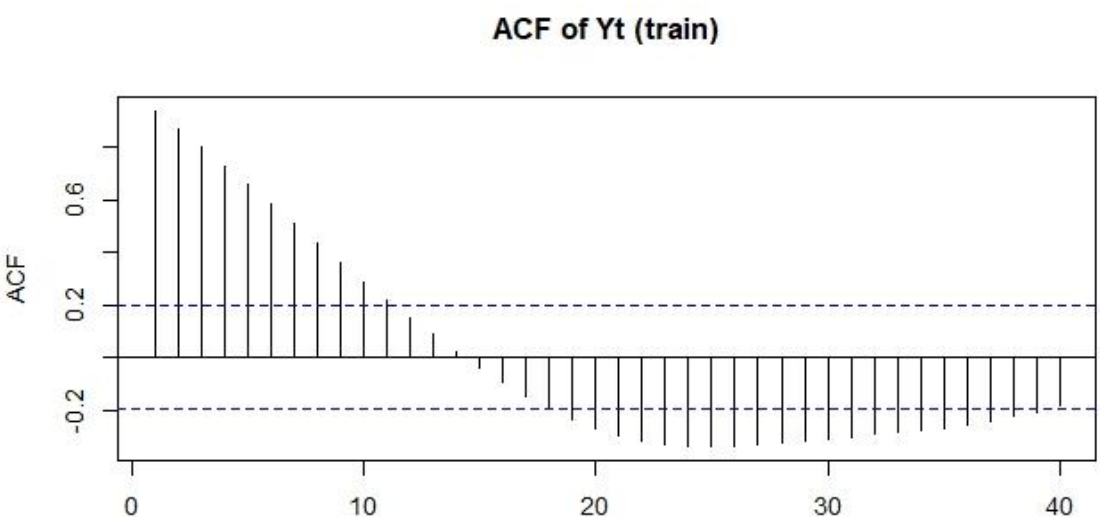


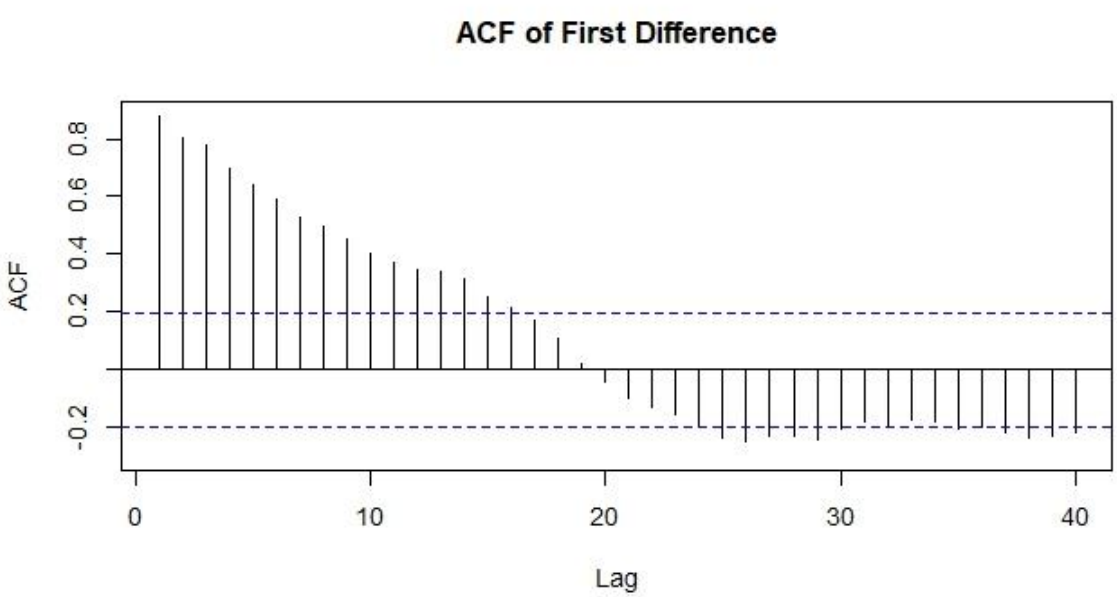
There is a gradual downward trend then there is a rapid upward trend.
Additionally, this time series does not fluctuate around a constant mean. Initially, the series remains close to zero, followed by a gradual decline reaching a minimum value around - 40. After this point, the series exhibits a sharp upward trend. The presence of a changing mean and strong trend suggests that the series may be non-stationary**.** Furthermore, there is no clear evidence of seasonal patterns or cyclic pattern in the data.

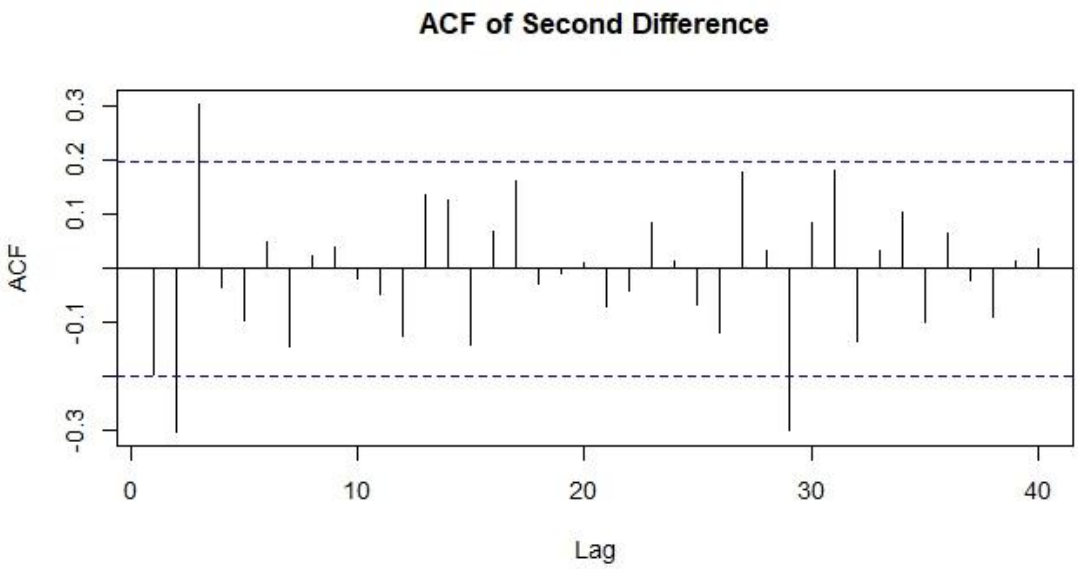
2. Analysis of d

(a) Display the autocorrelation function of (Yt) and $\nabla^k Y_t$, k = 1, 2 where $\nabla Y_t = Y_t - Y_{t-1}$. Interpret the results:
(1) What do you observe?

```
# Plot the autocorrelation function
Acf(ts_train, main="ACF of Yt (train)", lag.max =40)
# 1st difference
d1 <- diff(train)
Acf(d1, main="ACF of First Difference", lag.max =40)
# 2nd difference
d2 <- diff(train, differences = 2)
Acf(d2, main="ACF of Second Difference", lag.max =40 )
```







The ACF of the original series shows large positive autocorrelations that decline slowly across many lags. After applying the first difference, the autocorrelations still display gradual decay. For the second differenced series, most autocorrelations fall within the confidence bounds, indicating much weaker serial dependence.

(2) Using this informal graphical procedure, is there some evidence that the series (Yt) might be non-stationary? Explain carefully

Yes, there is evidence that the series is non-stationary.
The ACF plots of the original series and first difference exhibit slow decay pattern, with significant autocorrelations persisting across many lags. Such behaviour is characteristic of non-stationary time series with a possible unit root. When considering the second difference, it can be observed that most of the autocorrelations fall within the confidence bounds and decay quickly. This suggests that differencing helps remove the non-stationarity and that the series becomes rather stationary after differencing.

(b) To provide more evidence, implement a unit root test, the so-called Augmented DickeyFuller test (ADF), to assess whether there exists a root on the unit circle in the autoregressive lag polynomial of an AR(p) specification.

```
library(tseries)

adf_level <- adf.test(ts_train)
adf_level

adf_d1 <- adf.test(d1)
adf_d1

adf_d2 <- adf.test(d2)
adf_d2
```

```
>In adf.test(ts_train) : p-value greater than printed p-value
> adf_level

Augmented Dickey-Fuller Test
data:  ts_train
Dickey-Fuller = 0.023493, Lag order = 4, p-value = 0.99
alternative hypothesis: stationary

> adf_d1 <- adf.test(d1)
> adf_d1

Augmented Dickey-Fuller Test
data:  d1
Dickey-Fuller = -1.3683, Lag order = 4, p-value = 0.8381
alternative hypothesis: stationary

> adf_d2 <- adf.test(d2)
Warning message:
In adf.test(d2) : p-value smaller than printed p-value

> adf_d2
Augmented Dickey-Fuller Test
data:  d2
Dickey-Fuller = -4.5206, Lag order = 4, p-value = 0.01
alternative hypothesis: stationary
```

The Augmented Dickey–Fuller test applied to the original series yields a p-value of 0.99. Since this value is greater than the 5% significance level, the null hypothesis of a unit root cannot be rejected. Therefore, the original series is non-stationary.

The ADF test for the first differenced series produces a p-value of 0.838. Since the p-value is greater than 0.05, the null hypothesis of a unit root cannot be rejected, suggesting that the series remains non-stationary after first differencing.

For the second differenced series, the ADF test gives a p-value of 0.01. Since this value is less than the 5% significance level, the null hypothesis of a unit root is rejected. This indicates that the second differenced series is stationary.

Overall, the ADF test results confirm that the original series and the first differenced series are non-stationary, while the second differenced series is stationary. These results provide further evidence that differencing is required to achieve stationarity.

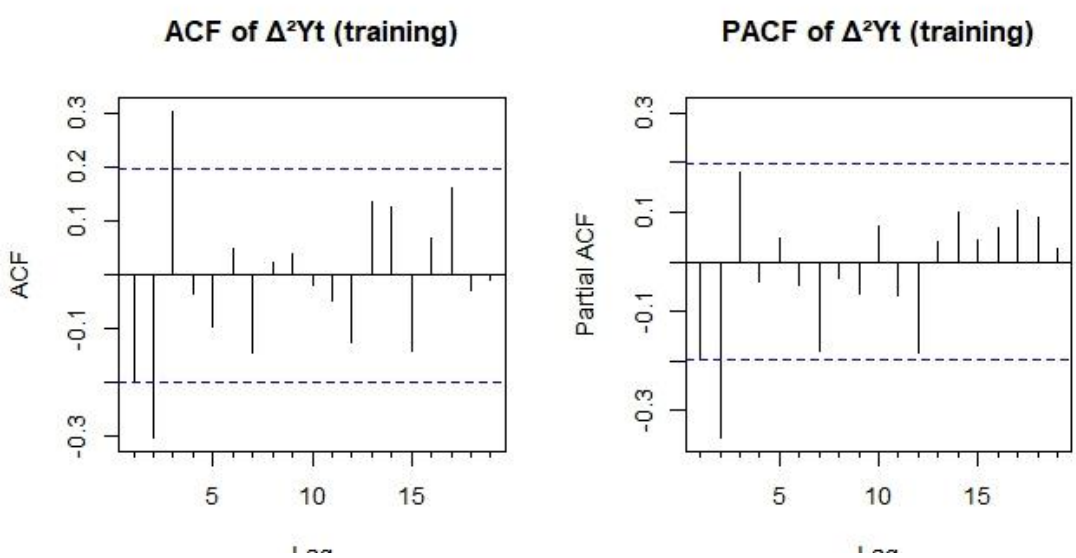
(c) Which order do you suggest for d?

d =2

Based on the results of the ACF analysis and the Augmented Dickey–Fuller (ADF) test, the original series and the first differenced series appear to be non-stationary. However, the second differenced series shows characteristics of stationarity. Therefore, the appropriate differencing order is **d = 2**, and the second differenced series will be used for further analysis.

3. Analysis of (p, q):
Using the result of part 2, plot the ACF and PACF of the relevant series (Yt or ∇kYt). Provide some upper bounds for p and q, say pmax and qmax.

```
d2 <- diff(train, differences = 2)

par(mfrow=c(1,2))   # ACF and PACF side-by-side
Acf(d2, main="ACF of Δ²Yt (training)")
Pacf(d2, main="PACF of Δ²Yt (training)")
par(mfrow=c(1,1))
```



In PACF plot, Lag 1 is strongly negative (definitely significant), Lag 2 also looks negative and large (and likely significant). But, after lag 2, the spikes are much smaller and mostly within bounds.
Therefore, **P max=2**

In ACF plot, Lag 1 is negative (just beyond the lower bound), Lag 2 is strongly negative, Lag 3 is strongly positive and after about lag 3, most spikes are inside the confidence bands.
There for **q max = 3**

Step 2: Estimation and Selection of ARIMA Models

1. Compute the AIC (Akaike's information criterion) and BIC (Bayesian information criterion) for the 16 remaining models

```
m11 <- Arima(train, order=c(1,2,1), include.constant=FALSE)
m12 <- Arima(train, order=c(1,2,2), include.constant=FALSE)
m13 <- Arima(train, order=c(1,2,3), include.constant=FALSE)
m14 <- Arima(train, order=c(1,2,4), include.constant=FALSE)
m21 <- Arima(train, order=c(2,2,1), include.constant=FALSE)
m22 <- Arima(train, order=c(2,2,2), include.constant=FALSE)
m23 <- Arima(train, order=c(2,2,3), include.constant=FALSE)
m24 <- Arima(train, order=c(2,2,4), include.constant=FALSE)
m31 <- Arima(train, order=c(3,2,1), include.constant=FALSE)
m32 <- Arima(train, order=c(3,2,2), include.constant=FALSE)
m33 <- Arima(train, order=c(3,2,3), include.constant=FALSE)
m34 <- Arima(train, order=c(3,2,4), include.constant=FALSE)
m41 <- Arima(train, order=c(4,2,1), include.constant=FALSE)
m42 <- Arima(train, order=c(4,2,2), include.constant=FALSE)
m43 <- Arima(train, order=c(4,2,3), include.constant=FALSE)
m44 <- Arima(train, order=c(4,2,4), include.constant=FALSE)
models <- list(m11,m12,m13,m14,m21,m22,m23,m24,m31,m32,m33,m34,m41,m42,m43,m44)
names(models) <- c("1,2,1","1,2,2","1,2,3","1,2,4",
                   "2,2,1","2,2,2","2,2,3","2,2,4",
                   "3,2,1","3,2,2","3,2,3","3,2,4",
                   "4,2,1","4,2,2","4,2,3","4,2,4")
results <- data.frame(
  Model = names(models),
  AIC = sapply(models, AIC),
  BIC = sapply(models, BIC)
)
results[order(results$AIC), ]
results[order(results$BIC), ]
```

```
> results[order(results$AIC), ]
      Model      AIC      BIC
2,2,1 2,2,1 291.3100 301.6499
3,2,1 3,2,1 293.0099 305.9347
2,2,2 2,2,2 293.0176 305.9425
4,2,2 4,2,2 293.3192 311.4139
4,2,1 4,2,1 294.7377 310.2475
1,2,3 1,2,3 294.7782 307.7031
2,2,3 2,2,3 294.9359 310.4457
3,2,2 3,2,2 294.9770 310.4868
1,2,4 1,2,4 295.3170 310.8268
4,2,3 4,2,3 295.6586 316.3383
2,2,4 2,2,4 295.7049 313.7997
1,2,2 1,2,2 295.9510 306.2909
3,2,3 3,2,3 296.9262 315.0210
3,2,4 3,2,4 297.4474 318.1271
4,2,4 4,2,4 297.5804 320.8451
1,2,1 1,2,1 301.1257 308.8806
```

```
> results[order(results$BIC), ]
      Model      AIC      BIC
2,2,1 2,2,1 291.3100 301.6499
3,2,1 3,2,1 293.0099 305.9347
2,2,2 2,2,2 293.0176 305.9425
1,2,2 1,2,2 295.9510 306.2909
1,2,3 1,2,3 294.7782 307.7031
1,2,1 1,2,1 301.1257 308.8806
4,2,1 4,2,1 294.7377 310.2475
2,2,3 2,2,3 294.9359 310.4457
3,2,2 3,2,2 294.9770 310.4868
1,2,4 1,2,4 295.3170 310.8268
4,2,2 4,2,2 293.3192 311.4139
2,2,4 2,2,4 295.7049 313.7997
3,2,3 3,2,3 296.9262 315.0210
4,2,3 4,2,3 295.6586 316.3383
3,2,4 3,2,4 297.4474 318.1271
4,2,4 4,2,4 297.5804 320.8451
```

I estimated 16 competing ARIMA (p, 2, q) models with p, q =1 to 4 using the training sample data. The AIC and BIC were computed for each model. Both criteria identify ARIMA (2,2,1) as the preferred specification since it yields the lowest AIC (291.31) and BIC (301.6499) values.

2. Provide a table with the estimates of your best three specifications.

summary(Arima(train, order=c(2,2,1)))
summary(Arima(train, order=c(3,2,1)))
summary(Arima(train, order=c(2,2,2)))

```
> summary(Arima(train, order=c(2,2,1)))
Series: train
ARIMA(2,2,1)

Coefficients:
          ar1      ar2     ma1
      -0.6248  -0.4263  0.4322
s.e.   0.1719   0.0904  0.1783

sigma^2 = 1.083:  log likelihood = -141.65
AIC=291.31   AICc=291.74   BIC=301.65

Training set error measures:
                      ME    RMSE      MAE       MPE     MAPE      MASE        ACF1
Training set 0.08745675 1.01449 0.838644 -157.8453 175.7976 0.4348626 0.002641309
```

```
> summary(Arima(train, order=c(3,2,1)))
Series: train
ARIMA(3,2,1)

Coefficients:
          ar1      ar2     ar3     ma1
      -0.4159  -0.3560  0.1118  0.2357
s.e.   0.4935   0.1624  0.2125  0.4927

sigma^2 = 1.091:  log likelihood = -141.5
AIC=293.01   AICc=293.66   BIC=305.93

Training set error measures:
                      ME     RMSE       MAE       MPE     MAPE      MASE         ACF1
Training set 0.08287425 1.012868 0.8351804 -152.3353 170.4405 0.4330666 -0.009154517
```

```
> summary(Arima(train, order=c(2,2,2)))
Series: train
ARIMA(2,2,2)

Coefficients:
          ar1      ar2     ma1     ma2
      -0.6820  -0.5152  0.5034  0.1079
s.e.   0.2086   0.1775  0.2450  0.1996

sigma^2 = 1.091:  log likelihood = -141.51
AIC=293.02   AICc=293.67   BIC=305.94

Training set error measures:
                     ME     RMSE       MAE       MPE     MAPE      MASE         ACF1
Training set 0.0836792 1.012899 0.8343266 -154.2676 172.3613 0.4326239 -0.009782555
```

Based on the AIC and BIC results in part 1, as well as the estimated coefficients, the models **ARIMA(2,2,1)**, **ARIMA(3,2,1)**, and **ARIMA(2,2,2)** are selected as the three best candidate specifications. These models have the lowest AIC and BIC values and are therefore considered suitable for further comparison and diagnostic analysis.

Step 3: Diagnostic tests

1. Among the different tests of specifications, one would like to test the absence of correlations of the error(residual) terms. Using the results of part 2 in Step 2, conduct a LjungBox test for the first 10 lags. Also, provide ACF and PACF plots. What can you conclude?

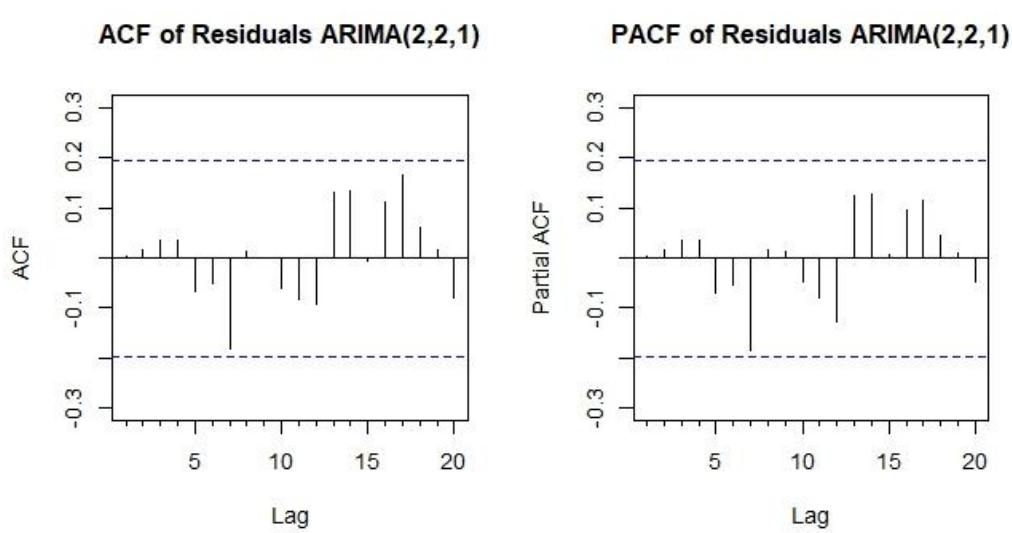
```
# fitting 3 models
mod1 <- arima(train, order=c(2,2,1))
res1 <- residuals(mod1)
mod1
mod2 <- arima(train, order=c(3,2,1))
res2 <- residuals(mod2)
mod2
mod3 <- arima(train, order=c(2,2,2))
res3 <- residuals(mod3)
mod3
# Jung-Box test for the first 10 lags
Box.test(res1, lag=10, type="Ljung-Box", fitdf=3)  # ARIMA(2,2,1)
Box.test(res2, lag=10, type="Ljung-Box", fitdf=4)  # ARIMA(3,2,1)
Box.test(res3, lag=10, type="Ljung-Box", fitdf=4)  # ARIMA(2,2,2)
# ACF and PACF of model
par(mfrow=c(1,2))
Acf(res1, main="ACF of Residuals ARIMA(2,2,1)")
Pacf(res1, main="PACF of Residuals ARIMA(2,2,1)")
par(mfrow=c(1,1))
#  ACF and PACF of mode2
par(mfrow=c(1,2))
Acf(res2, main="ACF of Residuals ARIMA(3,2,1)")
Pacf(res2, main="PACF of Residuals ARIMA(3,2,1)")
par(mfrow=c(1,1))
#  ACF and PACF of mode3
par(mfrow=c(1,2))
Acf(res3, main="ACF of Residuals ARIMA(2,2,2)")
Pacf(res3, main="PACF of Residuals ARIMA(2,2,2)")
par(mfrow=c(1,1))
```

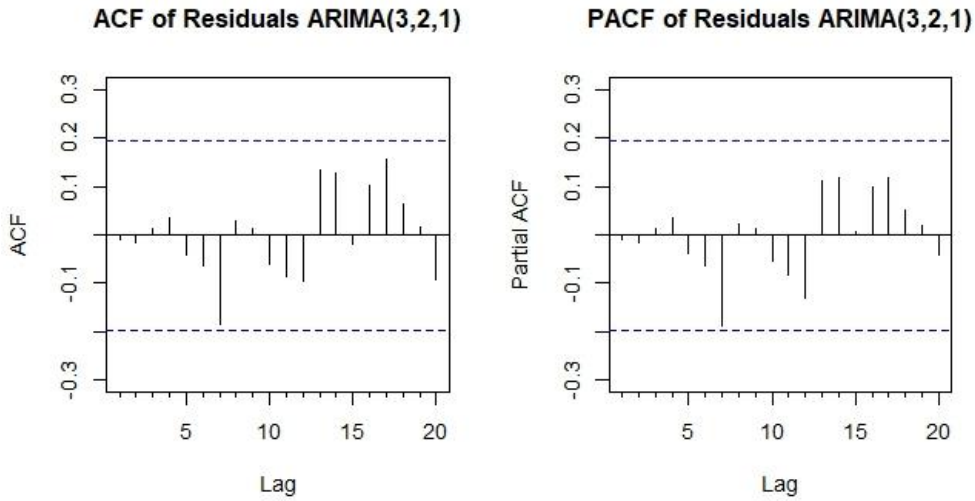
```
# residual plots
par(mfrow=c(3,1))
plot(res1, type="p", main="ARIMA(2,2,1) Residuals")
abline(h=0)
plot(res2, type="p", main="ARIMA(3,2,1) Residuals")
abline(h=0)
plot(res3, type="p", main="ARIMA(2,2,2) Residuals")
abline(h=0)
par(mfrow=c(1,1))
```

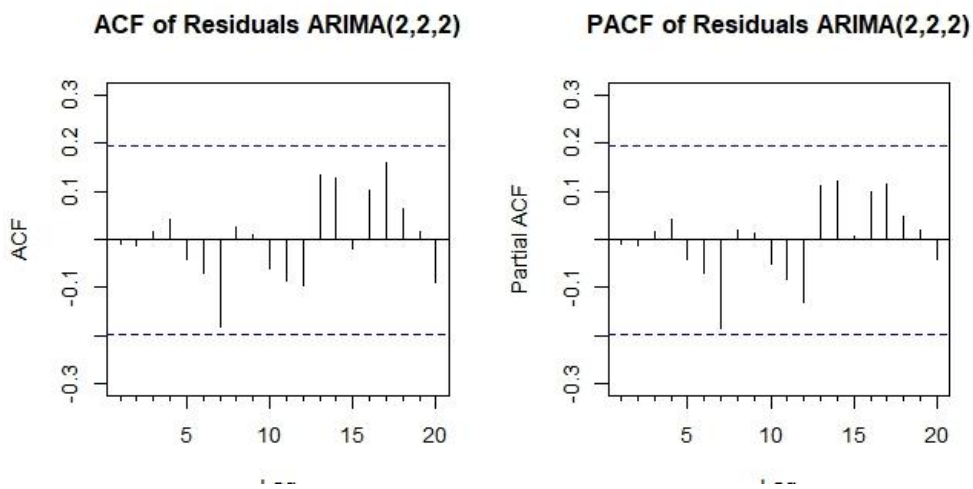
Ljung-Box test results

```
> Box.test(res1, lag=10, type="Ljung-Box", fitdf=3)  # ARIMA(2,2,1)

        Box-Ljung test

data:  res1
X-squared = 5.1174, df = 7, p-value = 0.6456

> Box.test(res2, lag=10, type="Ljung-Box", fitdf=4)  # ARIMA(3,2,1)

        Box-Ljung test

data:  res2
X-squared = 5.0536, df = 6, p-value = 0.537

> Box.test(res3, lag=10, type="Ljung-Box", fitdf=4)  # ARIMA(2,2,2)

        Box-Ljung test

data:  res3
X-squared = 4.9879, df = 6, p-value = 0.5454
```

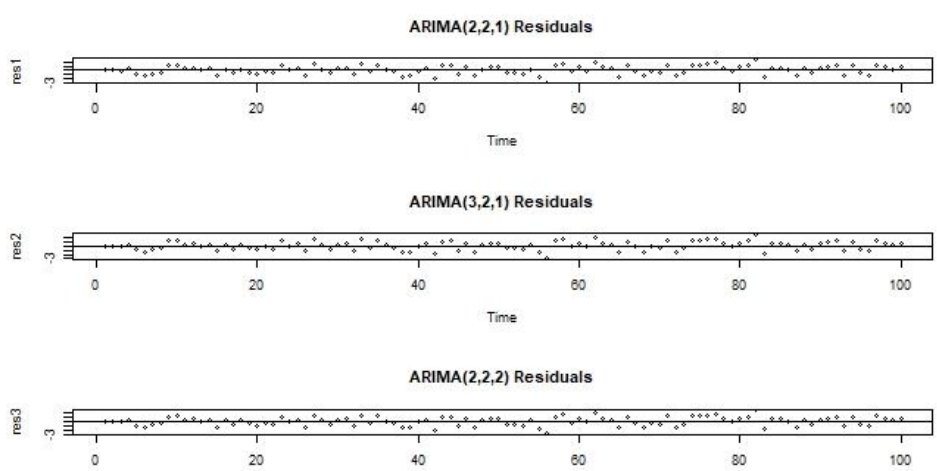
ACF and PACF plots







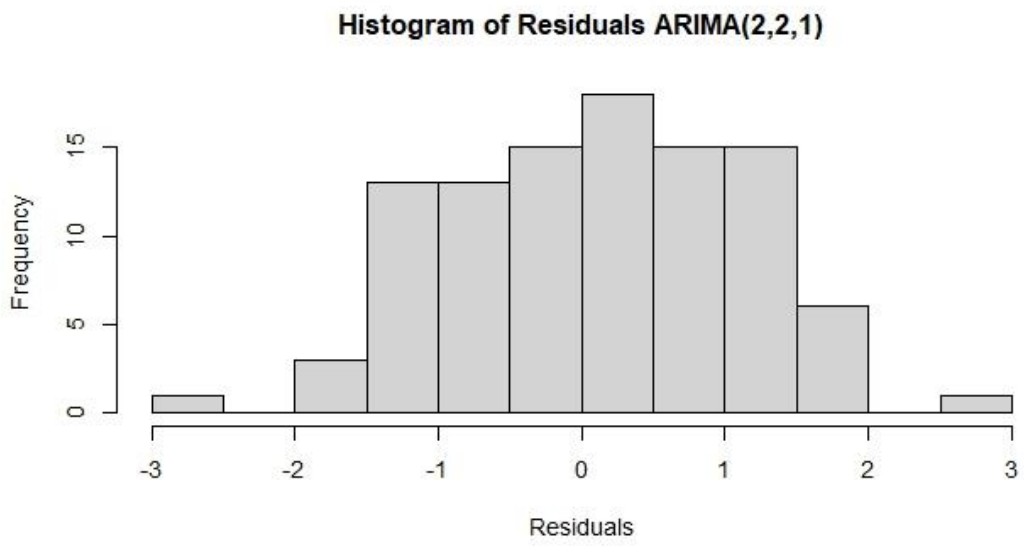
Residual Plots

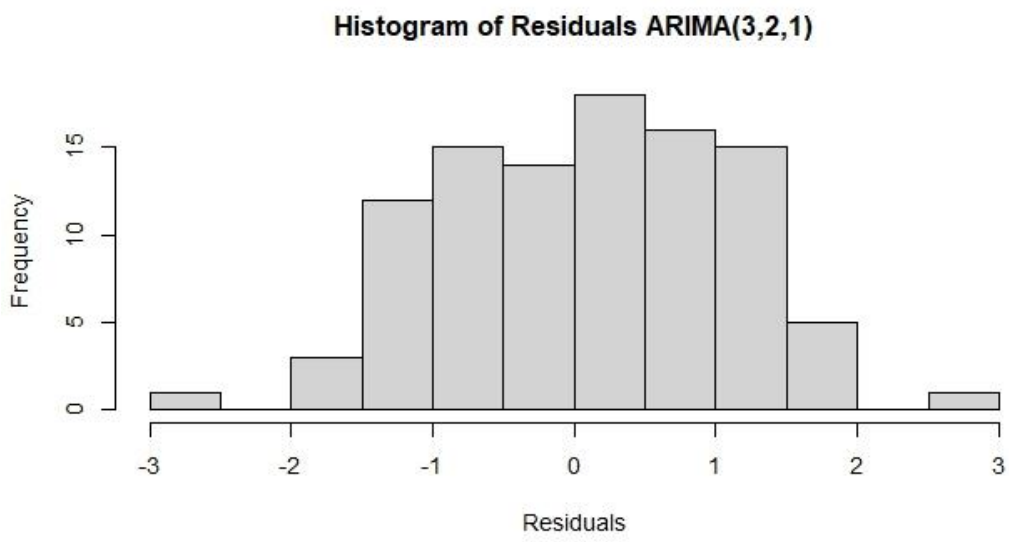


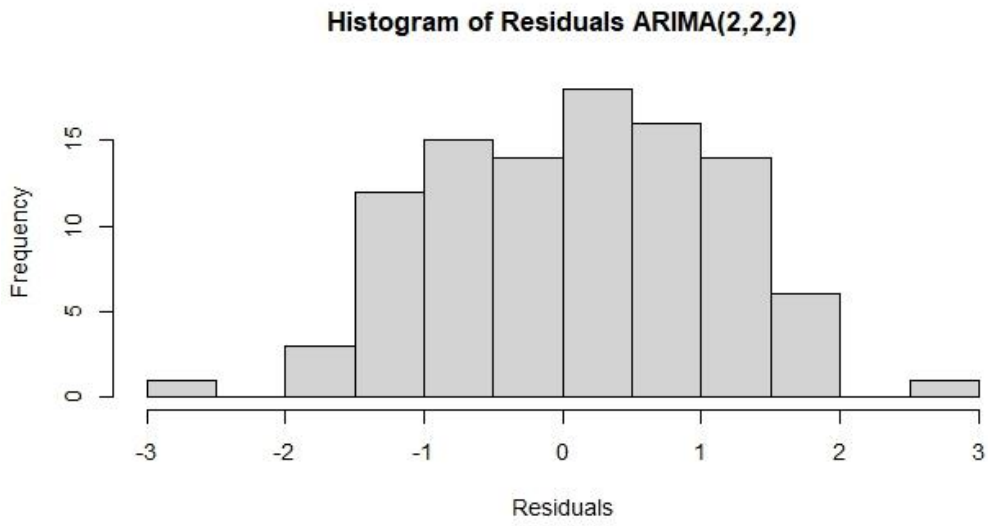
The ACF and PACF plots of the residuals for the three candidate models show that most autocorrelations lie within the confidence bounds, indicating no significant remaining serial correlation. Furthermore, the Ljung–Box test applied to the residuals with 10 lags produces p-values greater than 0.05 for all models, suggesting that the null hypothesis of no autocorrelation cannot be rejected. Therefore, the residuals behave approximately like white noise, indicating that the models adequately capture the dependence structure of the time series. Further, the residual plots also show that the residuals fluctuate randomly around zero with no visible pattern, indicating approximately constant variance and supporting the white-noise assumption.

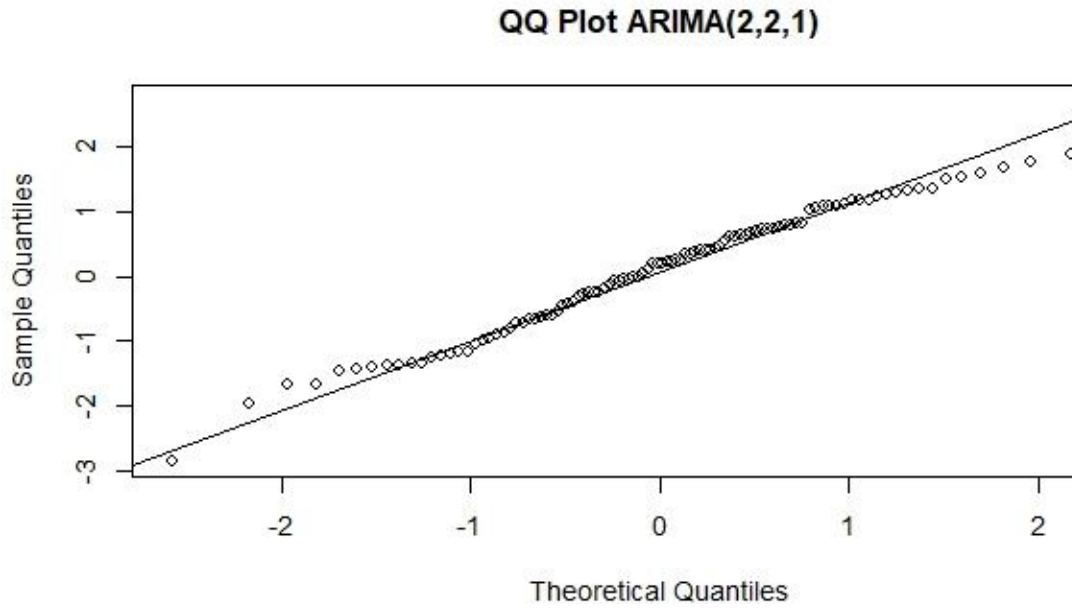
2. Are the residuals normal?
Hint: Use histograms, QQ plots, and Shapiro-Wilk's test to conclude the normality of the residuals for the selected 3 specifications.

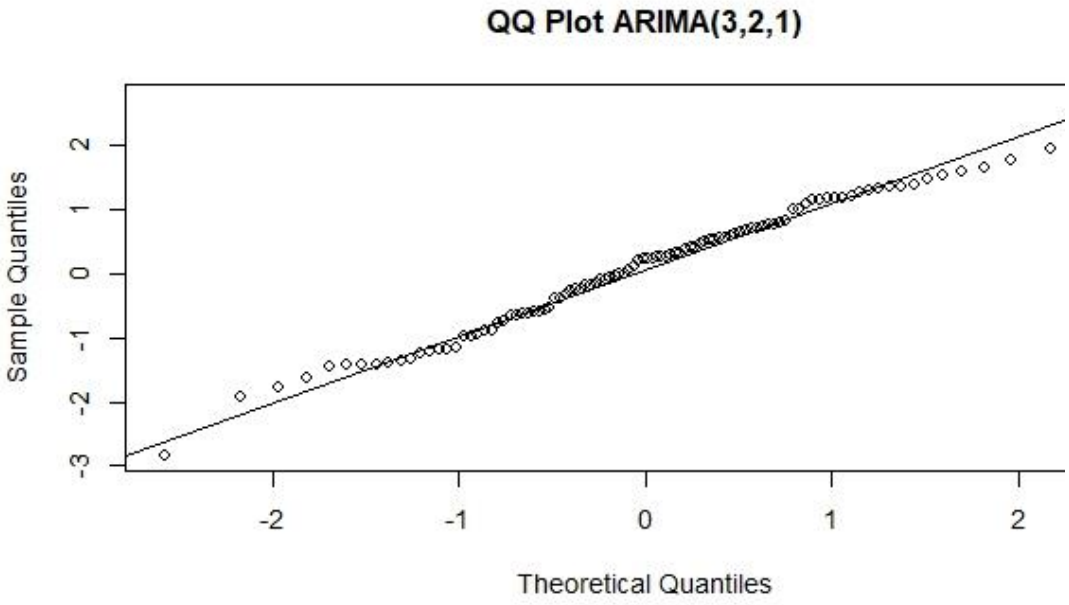
```
# Histograms
hist(res1, main="Histogram of Residuals ARIMA(2,2,1)", xlab="Residuals")
hist(res2, main="Histogram of Residuals ARIMA(3,2,1)", xlab="Residuals")
hist(res3, main="Histogram of Residuals ARIMA(2,2,2)", xlab="Residuals")
# QQ plot
qqnorm(res1, main="QQ Plot ARIMA(2,2,1)")
qqline(res1)
qqnorm(res2, main="QQ Plot ARIMA(3,2,1)")
qqline(res2)
qqnorm(res3, main="QQ Plot ARIMA(2,2,2)")
qqline(res3)
# Shapiro–Wilk test
shapiro.test(res1)
shapiro.test(res2)
shapiro.test(res3
```

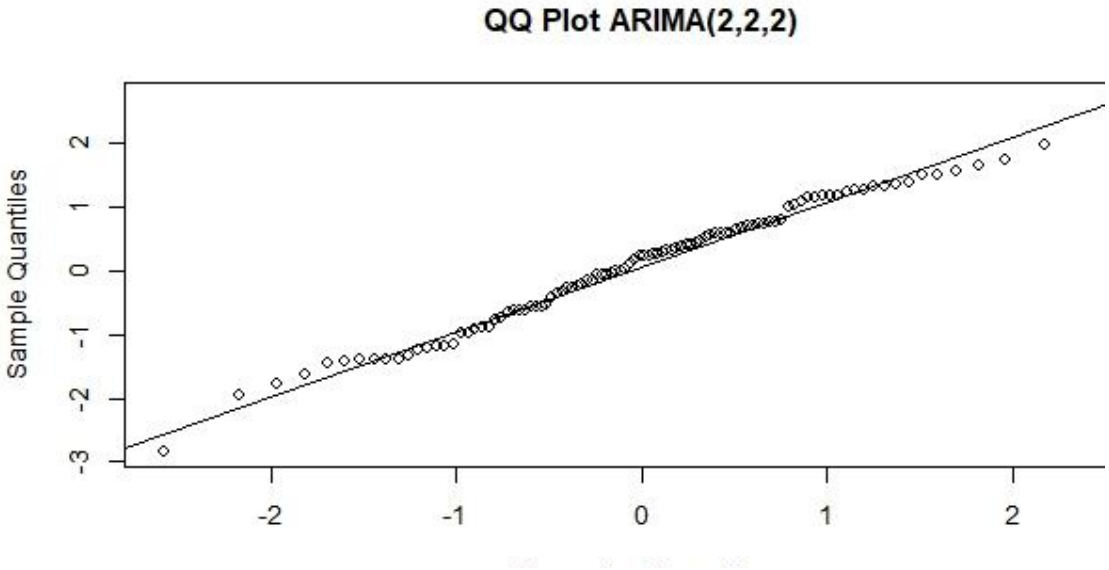





Histogram of Residuals ARIMA(2,2,2)
Frequency
0
5
10
15
-3
-2
-1
0
1
2
3
Residuals


QQ Plot ARIMA(2,2,1)
Sample Quantiles
-3
-2
-1
0
1
2
-2
-1
0
1
2
Theoretical Quantiles


QQ Plot ARIMA(3,2,1)
Sample Quantiles
Theoretical Quantiles
-3
-2
-1
0
1
2
-2
-1
0
1
2


QQ Plot ARIMA(2,2,2)
Sample Quantiles
-3
-2
-1
0
1
2
-2
-1
0
1
2

```
> shapiro.test(res1)

        Shapiro-Wilk normality test

data:  res1
W = 0.98745, p-value = 0.4683

> shapiro.test(res2)

        Shapiro-Wilk normality test

data:  res2
W = 0.9887, p-value = 0.5612

> shapiro.test(res3)

        Shapiro-Wilk normality test

data:  res3
W = 0.98857, p-value = 0.551
```

To assess the normality of the residuals, histograms, QQ plots and the Shapiro–Wilk test were applied to the residuals of the three ARIMA models. The histograms show approximately bell-shaped distributions, and the QQ plots indicate that most points lie close to the reference line, suggesting approximate normality. The Shapiro–Wilk test results further support this conclusion, as the p-values for all three models (0.4683, 0.5612, and 0.5510) are greater than 0.05. Therefore, we fail to reject the null hypothesis of normality, indicating that the residuals of the three ARIMA models can be considered approximately normally distributed.

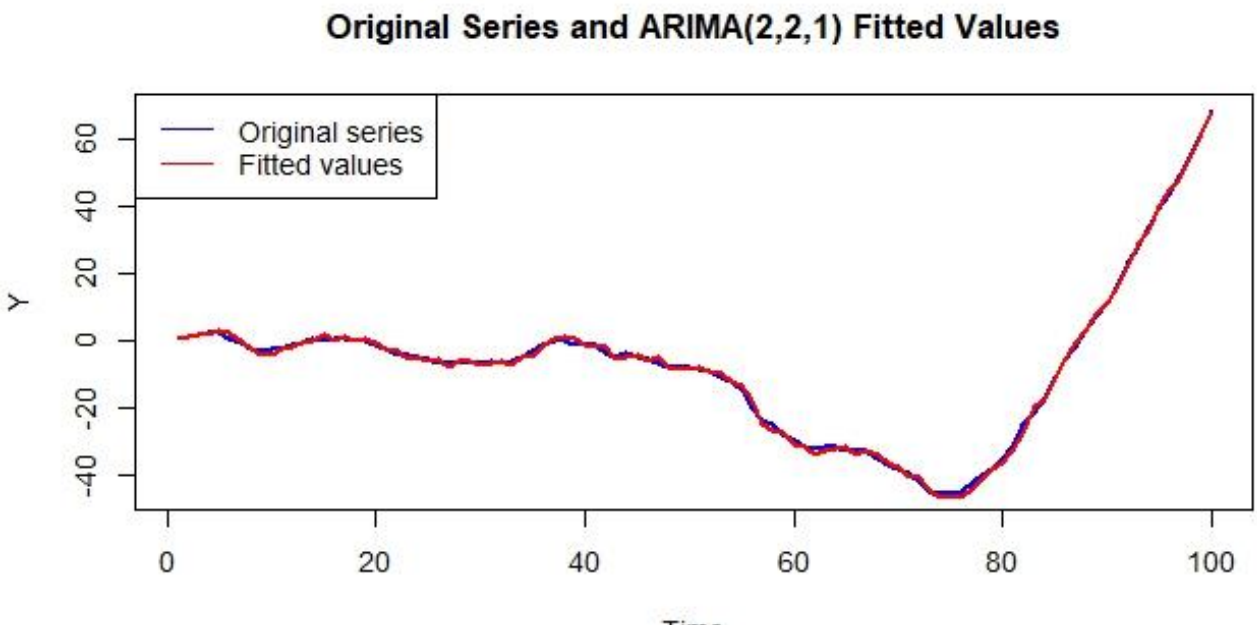
3. All in all, which model is your preferred specification?
Hint: Make use of all previous results as well as the principle of parsimony (i.e., among two competing models, one would generally select the one with less parameters, especially with respect to the moving average component).

best-fitted model: ARIMA (2,2,1)

Considering the results obtained in the previous steps, the three candidate models ARIMA(2,2,1), ARIMA(3,2,1), and ARIMA(2,2,2) all pass the diagnostic tests. The Ljung–Box test indicates no significant autocorrelation in the residuals, and the ACF and PACF plots confirm that the residuals behave like white noise. The histograms, QQ plots, and Shapiro–Wilk tests suggest that the residuals are approximately normally distributed. Among the competing models, ARIMA(2,2,1) provides the lowest AIC value. Furthermore, according to the principle of parsimony, the simplest model with fewer parameters is preferred. Therefore, ARIMA(2,2,1) is selected as the preferred specification.

4. Plot original time series and best-fitted model.

```
plot(train, type="l", col="blue",
    main="Original Series and ARIMA(2,2,1) Fitted Values",
    ylab="Y", xlab="Time", lwd=2)
lines(fitted(mod1), col="red", lwd=2)
legend("topleft",
      legend=c("Original series","Fitted values"),
      col=c("blue","red"),
      lty=1)
```

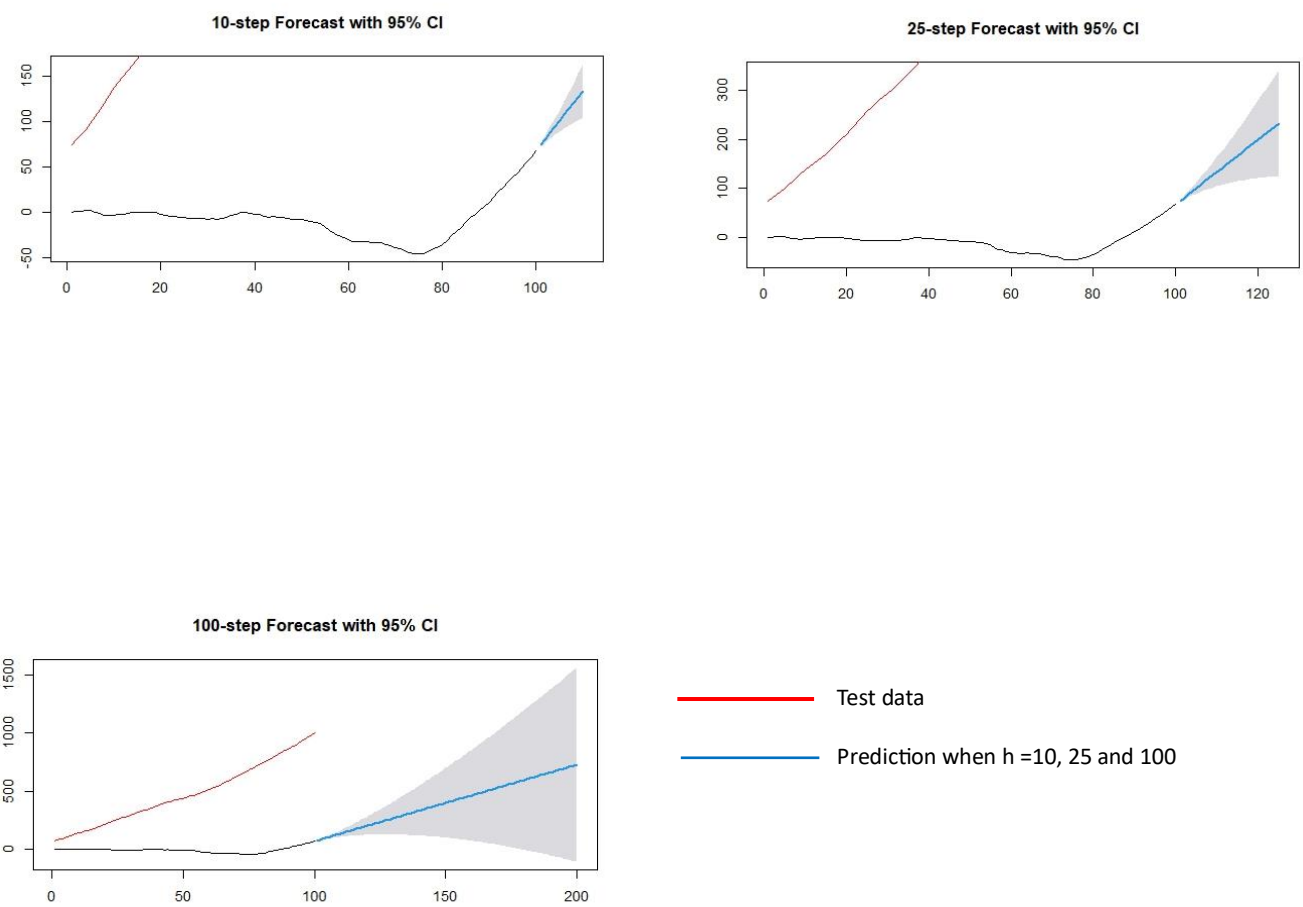


The fitted values closely follow the pattern of the original series, and the two lines overlap for most of the observations. This indicates that the ARIMA (2,2,1) model captures the underlying dynamics of the time series well. The close alignment between the original and fitted values suggests that the model provides a good in-sample fit and is able to explain the variations in the observed data.

Step 4: Forecast

1. Generate a forecast plot for a time series, incorporating a 95% confidence interval. Plot the original time series and h−step prediction. Take h = 10, h = 25 and h = 100.

```
fit <- Arima(train, order=c(2,2,1))
fc10  <- forecast(fit, h=10, level=95)  # when h=10
fc25  <- forecast(fit, h=25, level=95)  # when h=25
fc100 <- forecast(fit, h=100, level=95) # when h= 100
plot(fc10, main="10-step Forecast with 95% CI")   # when h=10
lines(test, col="red")
plot(fc25, main="25-step Forecast with 95% CI")   # when h=25
lines(test, col="red")
plot(fc100, main="100-step Forecast with 95% CI") # when h= 100
lines(test, col="red")
```



The forecast plot displays the original time series together with the predicted values obtained from the ARIMA(2,2,1) model. The grey shaded region represents the 95% confidence interval of the forecasts. The red line shows the actual out-of-sample observations, allowing a visual comparison between predicted and observed values. As the forecast horizon increases, the confidence intervals widen, indicating greater uncertainty in long-term forecasts.

2. Calculate MSE

```
mse10  <- mean((test[1:10]  - fc10$mean)^2)
mse25  <- mean((test[1:25]  - fc25$mean)^2)
mse100 <- mean((test[1:100] - fc100$mean)^2)  # same as mean((test - fc100$mean)^2)
mse10
mse25
mse100
```

```
When h = 10
> mse10
[1] 4.230966
```

```
When h = 25
> mse25
[1] 81.32036
```

```
When h = 100
> mse100
[1] 11387.39
```

The Mean Squared Error (MSE) was computed to evaluate the forecasting performance of the selected ARIMA (2,2,1) model. The MSE values increase as the forecast horizon increases: 4.23 for h=10, 81.32 for h=25, and 11387.39 for h=100. This indicates that the model provides relatively accurate short-term forecasts but becomes less accurate for longer horizons. The increase in MSE reflects the accumulation of forecast uncertainty and larger deviations between predicted and actual values in the long run.